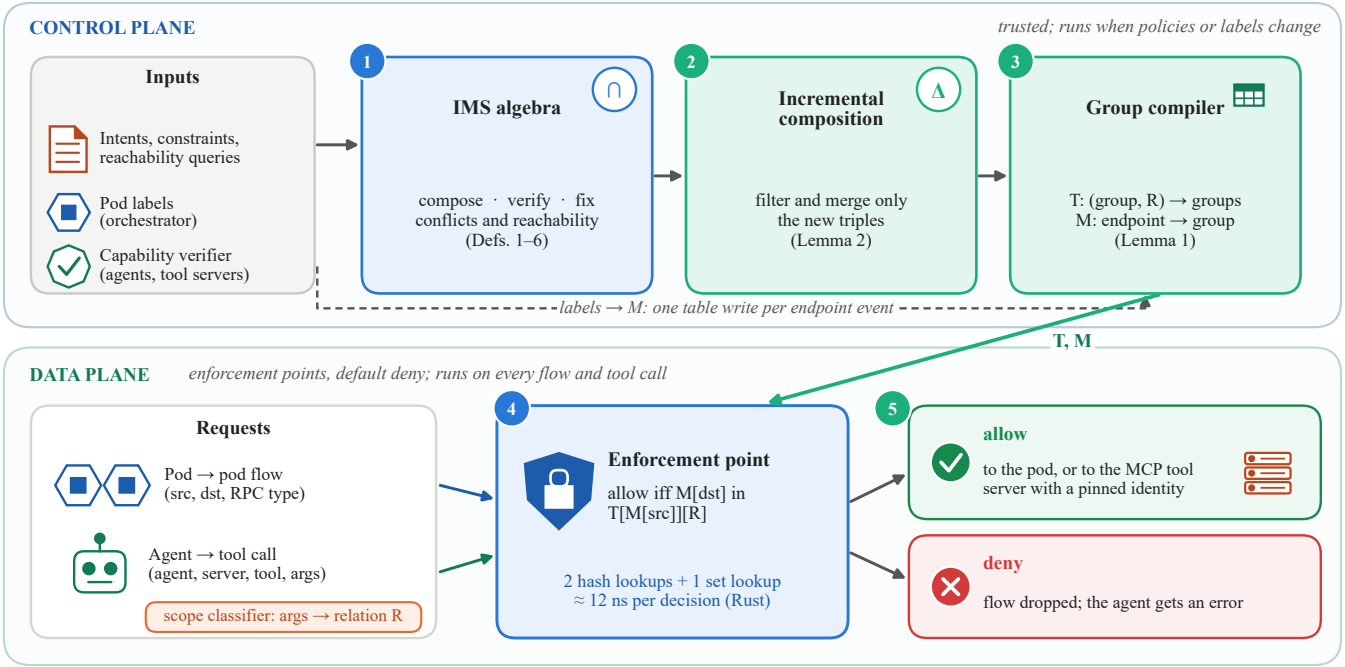

CONTROL PLANE
trusted; runs when policies or labels change
Inputs
Intents, constraints, reachability queries
Pod labels (orchestrator)
Capability verifier (agents, tool servers)
1
IMS algebra
compose · verify · fix conflicts and reachability (Defs. 1–6)
2
Incremental composition
filter and merge only the new triples (Lemma 2)
3
Group compiler
T: (group, R) → groups
M: endpoint → group
(Lemma 1)
labels → M: one table write per endpoint event
T, M
DATA PLANE
enforcement points, default deny; runs on every flow and tool call
Requests
Pod → pod flow (src, dst, RPC type)
Agent → tool call (agent, server, tool, args)
scope classifier: args → relation R
4
Enforcement point
allow iff M[dst] in T[M[src]][R]
2 hash lookups + 1 set lookup
≈ 12 ns per decision (Rust)
5
allow
to the pod, or to the MCP tool server with a pinned identity
deny
flow dropped; the agent gets an error